

AI Agents for Data Visualization and Kaggle Competitions

An end-to-end LLM agent (my-agent) benchmarked against two frozen yardsticks on 20 Kaggle competitions — with the apparatus needed to make the comparison honest

Wei-Hao Huang · Academia Sinica ISS Summer Internship 2026 · Report v4 (2026-08-17) · github.com/930731haohao-afk/kaggle-agent

1 · Motivation

Why these tasks are hard

- Kaggle competitions** demand the full expert loop — validation design, feature engineering, model selection, ensembling — where the difference between top entries sits at the 4th–5th decimal of the metric.
- Local cross-validation is an unreliable compass: it **contradicts the real leaderboard in 9 of 13** comparable competitions we measured.
- Data visualization** (EDA, diagnostics, reporting) is essential but repetitive: every dataset needs distribution, drift, and correlation views before modeling can be trusted.

Why an AI agent

- Tireless, systematic search: 22–80 candidate configurations per competition, every experiment logged and reproducible.
- Runs the same disciplined protocol on 20 competitions serially — a human cannot hold this consistency.
- Generates per-dataset EDA figures and per-competition reports automatically, so every claim is grounded in an artifact.

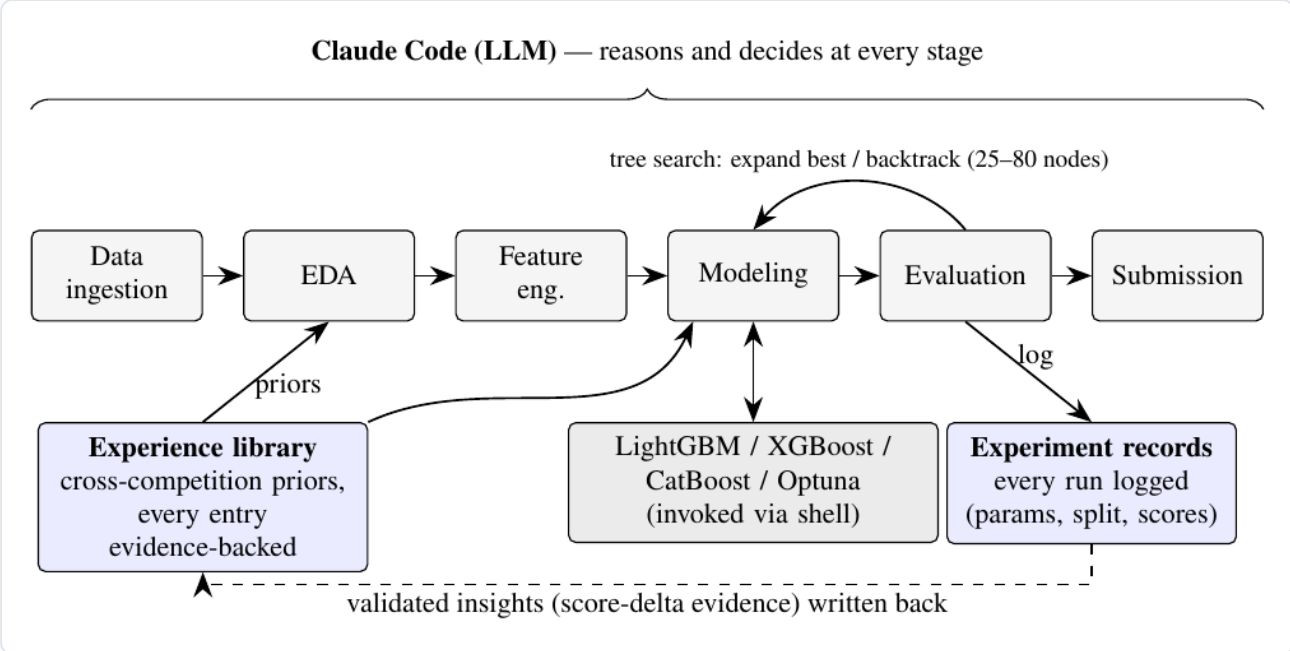
4 · Advantages & Difficulties

Expected → experienced

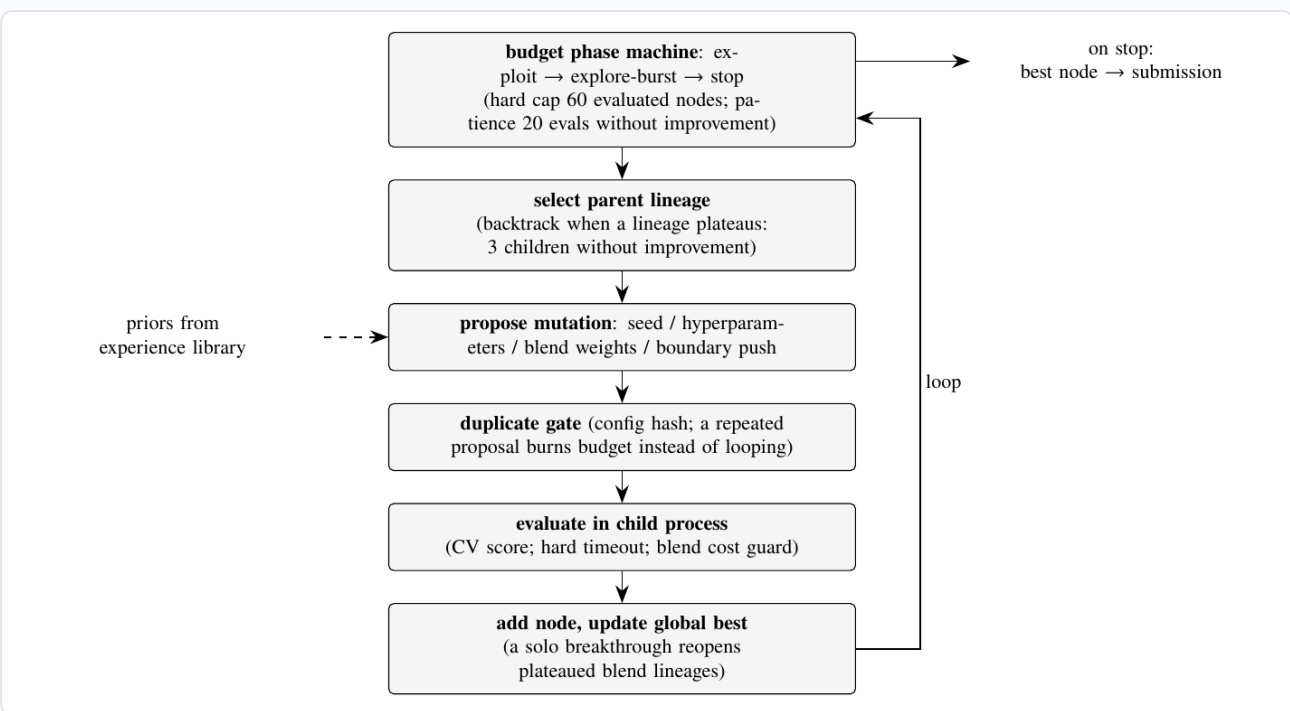
- Expected:** the agent would quickly beat most humans.
Experienced: median percentile 72.3 — solidly above average, with wins concentrated where no mature public solutions exist.
- Expected:** one run and one score decide a winner.
Experienced: score noise + run-to-run variance leave **27% of duels undecidable**; a paired error model became mandatory.
- Expected:** the agent follows the rules by default.
Experienced: a transcript audit caught one lane reading its own competition's recorded findings — and the clean re-run scored *worse*, proving the contamination advantage was real.
- Fairness is expensive: isolated data roots, credential-free sandboxes, and per-lane transcript audits cost real engineering time — but without them the numbers are not comparable.

2 · Workflow

An LLM reasons at every stage; Auto-ML libraries are invoked through the shell; a cross-competition **experience library** supplies evidence-backed priors; a **tree search** over candidate configurations is the optimisation loop.



Six-stage pipeline: problem dossier → EDA → features → modeling → evaluation → submission. The v5 dossier layer classifies the task and injects external data *upstream*, before any modeling.

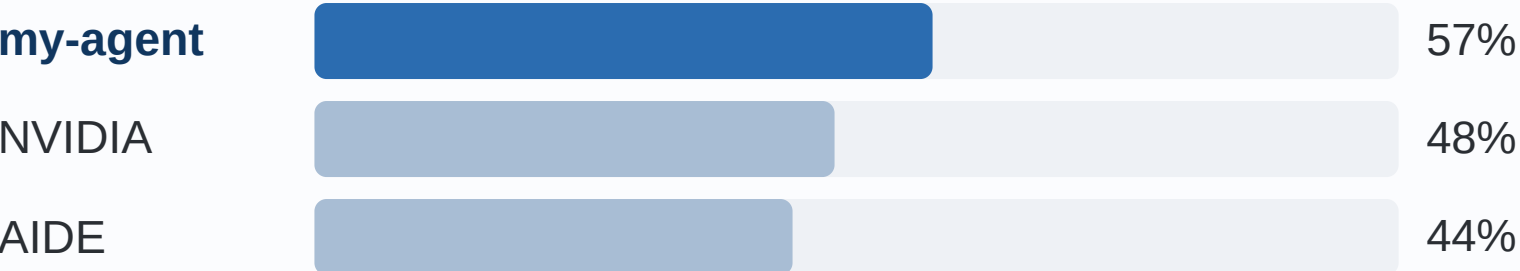


Tree-search loop: ensembles are first-class nodes with OOF caching and weight search; exploit until plateau, then one forced exploration burst.

3 · Results — Kaggle benchmark

All submissions scored on the real leaderboard (late submission); verdicts adjudicated by a paired test built from Kaggle's own public/private split.

Decidable-duel win rate (20 competitions — pre-v5 baseline runs)



27% of all duels are undecidable at the measurable error scale. Median percentile rank: my-agent 72.3, NVIDIA 68.0, AIDE 68.7. All standings here are the pre-v5 baseline lanes; the Stage-0.5 re-run below replaces them once scored.

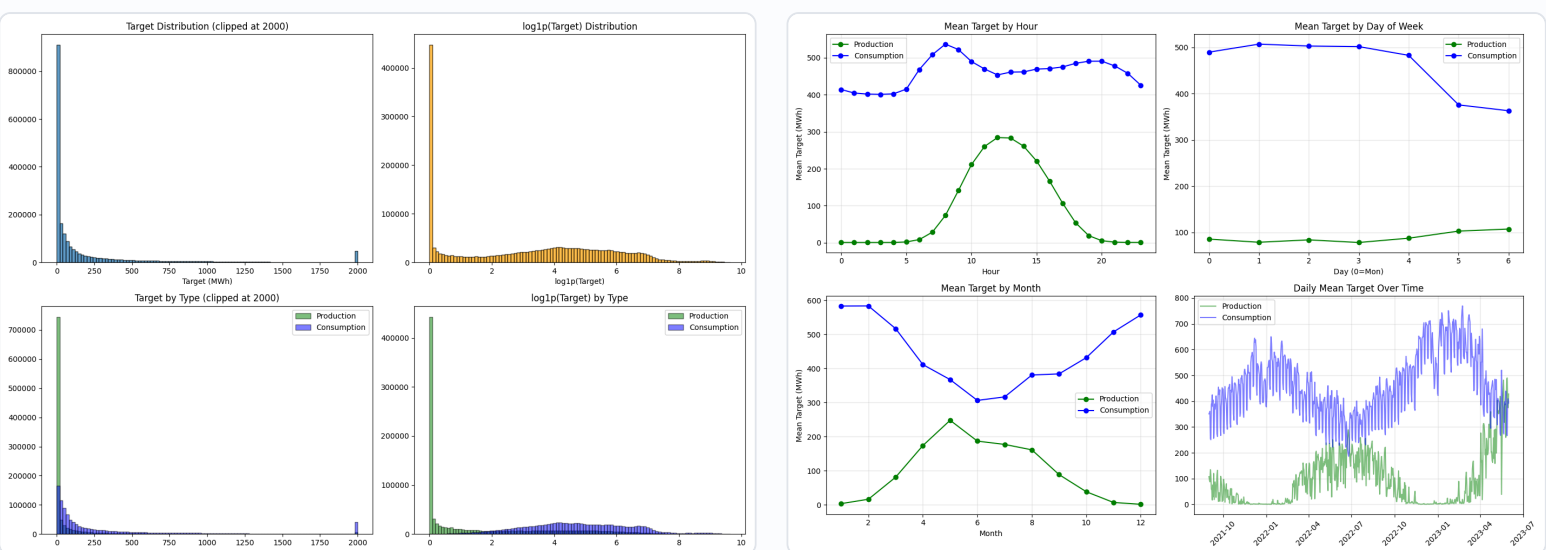
Mid-complexity three-way (offline-graded, private split)

Competition	Metric	my-agent	NVIDIA	AIDE
us-patent (NLP)	Pearson r ↑	0.86658	0.86160	0.65821
ventilator (time series)	MAE ↓	0.16525	0.40093	0.34591
SIIM-ISIC (imaging)	AUC ↑	0.93529	0.89822	0.88579

Final-architecture re-run (in progress, 2026-08-17)

All 20 competitions re-run under one frozen architecture, credential-free sandboxes, per-lane transcript audit. **15/20 lanes done**, 1 running, 4 queued — leaderboard scores **TBD** until the operator scoring step.

3b · Results — visualization output



Agent-generated EDA: target distribution and temporal-pattern diagnostics, produced automatically per dataset before modeling (example: energy-prosumer competition). The same pipeline emits per-competition ML-spec reports (22 published) and the flowcharts above.

5 · Discussion — advice

- State the error model with every claim.** A ranking without an error scale is noise: 27% of our duels are undecidable, and the winner flips when the evaluation set changes.
- Never trust local CV for between-agent claims.** Score on the real leaderboard; CV describes a pipeline's internal view only.
- Freeze your yardsticks.** Improving a baseline mid-study silently turns the benchmark into an optimisation of the baseline.
- Isolate mechanically, then audit the transcript.** Sandboxes stop leaks you anticipated; reading what the agent actually did catches the ones you didn't — in both directions (we hit one real contamination and one auditor false positive).
- Put knowledge injection upstream.** Injected at the blend stage it was worth $\sim 10^{-5}$; moved into problem identification it lifted the canonical failure case from last to first.

Where knowledge enters

- Downstream (blend stage): measured gain $\sim 10^{-5}$ — abandoned.
- Upstream (problem dossier): fires on exactly the 2/18 competitions that need external data; all **6 of 6** controlled external-data arms beat their matched baselines.
- But no local signal can select the *form* of injection — the one surviving selector died on its first pre-registered prediction.

Companion documents: Report v4 (Overleaf), Engineering Log, Case Studies, 22 per-competition ML-spec reports.